

Homework Assignment 5

Toy Model

All problems are graded on effort only.

Griffiths: 6.8, 6.12, 6.13, 6.15

Note on P6.8: There is a typo in the answer in some versions of the **digital** textbook for the course. The correct answer is:

$$\frac{\partial \sigma}{\partial \Omega} = \left(\frac{\hbar}{8\pi m_b c} \right)^2 |\mathcal{M}|^2$$

Hints on P6.15: This is a challenging problem, so I will give you plenty of help.

Let's use a consistent labeling to keep things clear below. Incoming particles: A has four-momentum p_a , B has p_b . Outgoing particles: A has p_1 and B has p_2 . The scattering angle θ is the angle between the incoming and outgoing A , so:

$$\vec{p}_a \cdot \vec{p}_1 = |\vec{p}_a| |\vec{p}_1| \cos \theta$$

P6.15a: The outgoing particles are not identical, so the second diagram does not come from crossing the outgoing particles (as in $A + A \rightarrow B + B$). This process still has two diagrams, one with the internal C line horizontal (we call this a s -channel process) and one with internal C line vertical (we call this a u -channel process). Your answer will look like this:

$$\mathcal{M} = g^2 \left[\frac{1}{(X - m_C^2 c^2)} + \frac{1}{(Y - m_C^2 c^2)} \right]$$

where X and Y are each one of the Mandelstom variables s, t , or u :

$$\begin{aligned} s &= (p_a + p_b)^2 \\ t &= (p_a - p_1)^2 \\ u &= (p_a - p_2)^2 \end{aligned}$$

P6.15b: Notice that now $\vec{p}_a = -\vec{p}_b$ and $\vec{p}_2 = -\vec{p}_1$. Also:

$$|\vec{p}_a| = |\vec{p}_b| = |\vec{p}_1| = |\vec{p}_2| \equiv p$$

First show that:

$$s = (p_a + p_b)^2 = 4E^2$$

and

$$u = (p_a - p_2)^2 = -2p^2 (1 + \cos \theta) = -4p^2 \cos^2 \left(\frac{\theta}{2} \right)$$

Make sure your answer for $d\sigma/d\Omega$ is a function of E and θ variables only. If you like, you can simplify a factor like:

$$\left(\frac{a}{b} + \frac{c}{d} \right)^2 \rightarrow \left(\frac{ad + cd}{bd} \right)^2$$

but it won't get that much prettier.

P6.15c: Use P6.8. Also show that in the limit of large m_B :

$$s = u = m_B^2 c^4$$

P6.15d: Easy integral! No θ or ϕ dependence!